

Mesh Convergence Study

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1 Overview

Using the power-law, velocity-driven LBM program, we will conduct a mesh convergence study for shear-thinning, shear-thickening and Newtonian fluids.

The mesh convergence study consists of running various, in our case 4, simulations of different mesh sizes with all the simulations having the same Reynolds number.

There is a relationship between each mesh size and number of iteration for that mesh. The starting mesh $M_1 = (N_x, N_y)$ is run for I_1 iterations. The following mesh $M_2 = (2N_x, 2N_y)$ is four times the size of M_1 and is run for $I_2 = 4I_1$ iterations. Following this reasoning, we have the n^{th} mesh $M_n = (nN_x, nN_y)$ is ran for $I_n = n^2I_1$ iterations. As mentioned, we will use $n = 4$ mesh sizes in this convergence study but it can be extended to more.

For this study we define the Reynolds number as,

$$\text{Re} = \frac{U_{max} N_y}{\nu_c} \quad (1)$$

where U_{max} is the maximum velocity of the fluid, N_y is the height of our mesh and ν_c is the apparent viscosity at the centreline which is found using

$$\nu_c = \frac{1}{3} \left(\tau_c - \frac{1}{2} \right) \quad (2)$$

where τ_c is the relaxation time at the centreline.

Moreover, since for consequent mesh sizes the size of N_y is doubled, we impose that the maximum fluid velocity U_{max} must be *halved*. Since our simulation takes inputs of n, m , we will find the value of m such that the maximum velocity is halved when our height N_y is doubled.

Recall that for a velocity-driven flow, the maximum velocity U_{max} has the expression,

$$U_{max} = \left[\frac{\rho}{m\text{Re}} N_y^{(2n-1)/n} \left(\frac{nN_y}{2(n+1)} \right)^{n-1} \right]^{1/(n-2)} \quad (3)$$

Rearranging this expression for m ,

$$m = \frac{\rho}{U_{max}^{n-2}\text{Re}} N_y^{(2n-1)/n} \left(\frac{nN_y}{2(n+1)} \right)^{n-1} \quad (4)$$

We now choose a Reynolds number and produce four mesh sizes with corresponding iterations and m for each mesh so that the maximum velocity halves as the mesh height doubles.

Once the simulations are ran, we can explore the errors. The LBM is of second order accuracy, for each simulation with corresponding mesh size, we calculate the error ϵ ,

$$\epsilon = \sqrt{\frac{\sum_{k=1}^N \|\mathbf{u}^{\text{num}}(k) - \mathbf{u}^{\text{ana}}(k)\|^2}{\sum_{k=1}^N \|\mathbf{u}^{\text{ana}}(k)\|^2}} \quad (5)$$

with the L^2 norm $\|\cdot\|$, analytic velocity field $\mathbf{u}^{\text{ana}}$, numerical velocity $\mathbf{u}^{\text{num}}$, site k , number of lattices $N = N_x \times N_y$ and plot $\log(\epsilon)$ on the y -axis and $\log(N_y)$ on the x -axis. Since we have second order accuracy for the LBM, we *expect* a straight line with gradient of -2 .

2 Small Reynolds Number

Here, we perform a mesh convergence study for power-law fluids with flow behaviour indices of $n = 0.7$, 1.0 , and 1.3 , representing shear-thinning, Newtonian, and shear-thickening behaviour, respectively, at a small Reynolds number ($\text{Re} = 1.0$).

2.1 Shear-Thinning Fluid

For a shear-thinning fluid with $n = 0.7$, we have the following results.

Mesh	N_y	Iterations	U_{max}	m	$\log(\epsilon)$	$\log(N)$
M_1	8	25,000	4.50×10^{-2}	5.00×10^{-2}	-3.37	2.08
M_2	16	100,000	2.25×10^{-2}	2.45×10^{-2}	-4.72	2.77
M_3	32	400,000	1.12×10^{-2}	1.20×10^{-2}	-5.89	3.47
M_4	64	1,600,000	5.61×10^{-3}	5.89×10^{-3}	-7.07	4.16

Table 1: Simulation parameters and results for $\text{Re} = 1.0$, $n = 0.7$.

2.2 Newtonian Fluid

For a Newtonian fluid with $n = 1.0$, we have the following results.

Mesh	N_y	Iterations	U_{max}	m	$\log(\epsilon)$	$\log(N)$
M_1	8	25,000	6.25×10^{-2}	1.00×10^{-1}	-5.03	2.08
M_2	16	100,000	3.13×10^{-2}	1.00×10^{-1}	-6.33	2.77
M_3	32	400,000	1.56×10^{-2}	1.00×10^{-1}	-7.69	3.47
M_4	64	1,600,000	7.81×10^{-3}	1.00×10^{-1}	-9.06	4.16

Table 2: Simulation parameters and results for $\text{Re} = 1.0$, $n = 1.0$.

2.3 Shear-Thickening Fluid

For a shear-thickening fluid with $n = 1.3$, we have the following results.

Mesh	N_y	Iterations	U_{max}	m	$\log(\epsilon)$	$\log(N)$
M_1	8	25,000	1.82×10^{-2}	1.00	-4.00	2.08
M_2	16	100,000	9.11×10^{-3}	1.78	-4.84	2.77
M_3	32	400,000	4.55×10^{-3}	3.16	-5.78	3.47
M_4	64	1,600,000	2.28×10^{-3}	5.63	-6.79	4.16

Table 3: Simulation parameters and results for $\text{Re} = 1.0$, $n = 1.3$.

2.4 Discussion

Following the simulations, we plot the log error on the y -axis and the log of the mesh size on the x -axis. We then find the gradient of the straight line of best fit (SLOBF) and compare it to the hypothesised gradient of -2 .

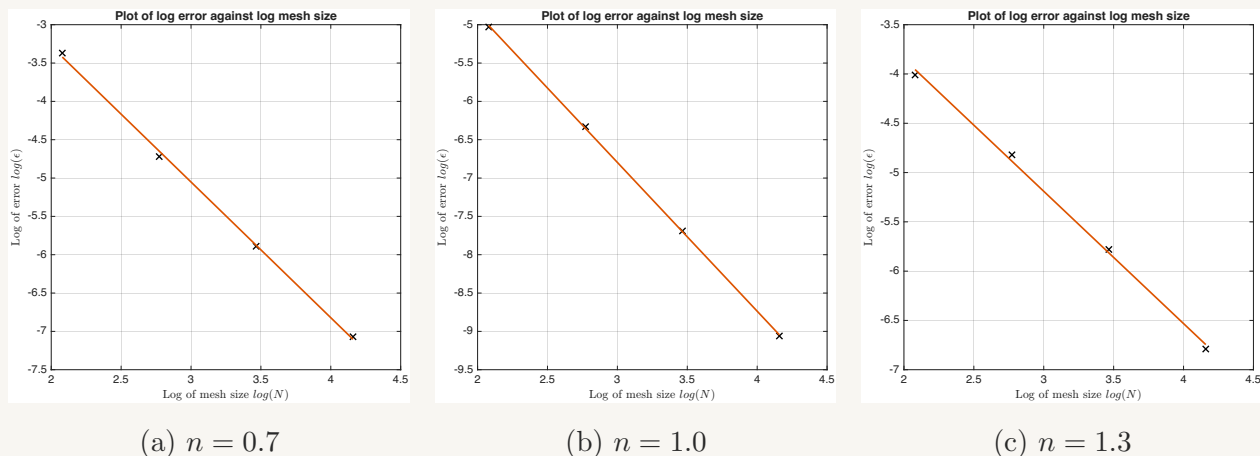


Figure 1: Plots for the fluids at $\text{Re} = 1.0$.

Flow behaviour index	Gradient of SLOBF	Percentage error
$n = 0.7$	-1.7702	11.49%
$n = 1.0$	-1.9404	2.98%
$n = 2.0$	-1.3417	32.92%

Table 4: Results for $\text{Re} = 1.0$.